

Trigger Output & Timing Analysis

Keysight EDU33210 Series — Phase Modulation Mode

Keysight Control UW

2026-03-30

1 Overview

This document describes the hardware trigger output in phase-modulation mode and quantifies the delay between the trigger signal and the stimulation pulse.

Central question: **What is the delay between the stimulation output and the trigger output?**

2 Current Implementation

The phase-mode protocol (`_run_phase_protocol`, `src/lib/stimulator.py`) **already emits a hardware trigger with every pulse**:

1. **Trigger output enabled** on the rear-panel Ext Trig connector (line 314): `:OUTPut1:TRIGger:STATe 1`
2. **Bus trigger per pulse** in the pulse-train loop (line 334): `*TRG` — simultaneously starts the burst on both channels and fires the trigger output.
3. **Trigger source set to BUS** (line 311): `:TRIGger{ch}:SOURce BUS` — keeps the Ext Trig connector free for output (it would be unavailable if set to `EXTernal`, since the same connector doubles as input).

2.1 SCPI Commands

Command	Purpose	Location
<code>:TRIGger{ch}:SOURce BUS</code>	Trigger source = software bus	<code>stimulator.py:311</code>
<code>:SOURce{ch}:BURSt:MODE TRIGgered</code>	Burst waits for trigger	<code>stimulator.py:155</code>
<code>:SOURce{ch}:BURSt:NCYCles {n}</code>	Finite burst length	<code>stimulator.py:153</code>
<code>:OUTPut1:TRIGger:STATe 1</code>	Enable trigger output	<code>stimulator.py:314</code>
<code>*TRG</code>	Fire bus trigger	<code>stimulator.py:334</code>

2.2 Not Currently Configured

Trigger slope/polarity is not explicitly set (device defaults to rising edge). Relevant command:
:OUTPut1:TRIGger:SLOPe POSitive

3 Signal Path & Timing

```
Python: dev.write('*TRG')
|
| [USB transfer: ~1-5 ms, variable]
v
Keysight receives SCPI command
|
| [Internal processing: < 135 ns]
v
Simultaneous hardware events:
+---> Ch1 burst output (0 deg phase)
+---> Ch2 burst output (180 deg phase)
+---> Ext Trig connector pulse (trigger out)
```

3.1 Delay Budget

Trigger output vs. stimulation pulse — both are generated by the same internal trigger event. They are hardware-synchronized:

Parameter	Value	Source
Internal trigger latency	< 135 ns	33500B datasheet (trigger delay = 0)
Internal trigger jitter	< 2.5 ns RMS	33500B datasheet
Trigger output pulse width	> 1 us (BUS mode)	33500 series user manual

USB communication latency — determines *when* the pulse fires relative to Python's clock, but does **not** affect trigger-to-stimulus synchronization (both originate from the same hardware event):

Parameter	Typical Value	Notes
USB write latency	1–5 ms	Python -> pyvisa -> USB -> device
USB latency jitter	~0.5–2 ms	OS scheduling + USB protocol

Software ITI jitter — `time.sleep()` jitter is 1–10 ms (OS-dependent), absorbed by the randomized ITI range (e.g. 900–1100 ms). Does not affect trigger-to-stimulus sync.

4 Trigger Output Electrical Specs

Specs from the 33500B datasheet. EDU33210 is architecturally similar; exact values may differ slightly.

Parameter	Value
Output voltage (open circuit)	3 Vpp nominal
Output voltage (50 ohm load)	1.5 Vpp nominal
Output impedance	50 ohm
Minimum pulse width	16 ns
Polarity	Rising (default) or falling edge
Connector	Rear-panel Ext Trig / Sync (BNC)

5 Connection

Keysight EDU33210 (rear panel) Receiving System
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Ext Trig / Sync [BNC] ----BNC----> Trigger Input
Ch1 Output [BNC] -----cable-----> Electrode 1
Ch2 Output [BNC] -----cable-----> Electrode 2

- For falling-edge detection: `:OUTPut1:TRIGger:SLOPe NEGative`
- Verify with an oscilloscope before running a real session: trigger edge should coincide with burst onset within < 1 us.

6 Current Logging Gaps

Gap	Recommendation
Timestamps at second resolution (%H:%M:%S)	Use <code>time.time()</code> / <code>time.perf_counter()</code> for ms precision
Individual pulses not in CSV	Add per-pulse rows with high-resolution timestamps
Trigger slope not explicitly set	Add <code>:OUTPut1:TRIGger:SLOPe POSitive</code> to setup
No trigger status printed for operator	Show trigger config at session start

7 Codebase Reference

Component	File	Lines
Phase device setup	<code>src/lib/stimulator.py</code>	148–161
Trigger output enable	<code>src/lib/stimulator.py</code>	314
Trigger source config	<code>src/lib/stimulator.py</code>	311
Per-pulse bus trigger	<code>src/lib/stimulator.py</code>	334
Pulse train loop	<code>src/lib/stimulator.py</code>	333–341
ERP stim config	<code>src/ERP_stim.py</code>	40–49
Session logger	<code>src/lib/logger.py</code>	89–124

8 References

- Keysight 33500B/33600A Series Datasheet (5992-2572)
- Keysight 33500 Series User's Guide (33500-90901)
- EDU33210 Series Datasheet (3121-1004)